Flow fields in microfluidic chips

Microfluidic chips move tiny volumes of fluid through channels that are narrower than a human hair. Efficient mixing is difficult in such channels because the flow is laminar and diffusion is slow. Engineers therefore add herringbone grooves or serpentine sections that fold the fluid onto itself. The resulting flow field is complicated, and characterization usually requires fluorescence microscopy.

In this note we describe an offline workflow that reconstructs the flow field from short video clips. The workflow first identifies fluorescent particles, then links them into trajectories across frames. A final step fits a smooth velocity profile to the trajectories with a regularized interpolation.

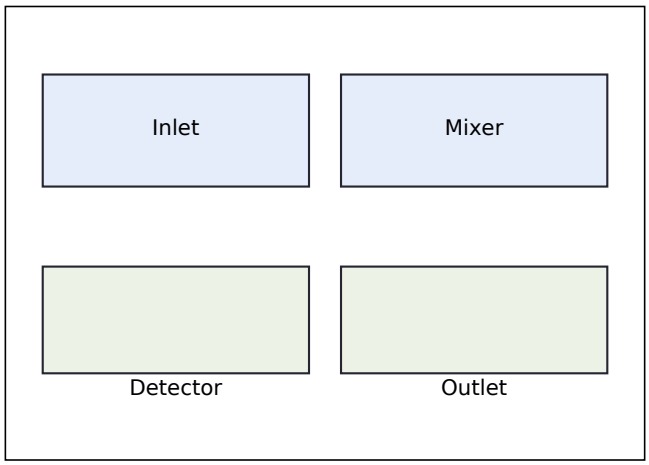

Figure 1: Layout of the test chip. Fluid enters at the inlet, passes the mixer and the detector, and leaves through the outlet.

Particle identification uses a difference of Gaussians filter followed by a local maximum search. We found that a fixed threshold suffices for most clips, but flickering illumination needs an adaptive threshold. Trajectory linking is the most fragile step, since particles leave the focal plane and reappear later. Our linker tolerates gaps of up to three frames and rejects links that imply unphysical accelerations. The interpolation step uses a thin plate spline with a stiffness coefficient chosen by cross validation. On synthetic benchmarks the reconstructed velocity field differs from the reference by less than four percent. The largest differences appear close to the channel walls, where few particles are observed. Increasing the particle density reduces this effect but also increases the number of ambiguous links. A practical compromise is a density of roughly one particle per fifty square micrometres. Finally, all intermediate files are stored so that every reconstruction can be audited and reproduced.

We tested the workflow on three chip designs with different groove configurations. Each design was filmed at five flow rates, giving fifteen clips of twenty seconds each. Processing one clip takes about four minutes on an ordinary office laptop. The differential pressure across the chip was recorded independently as a consistency check.

Table 1: Reconstruction error for each chip design.

Design	Grooves	Error	Links
Straight	none	2.1 %	4120
Herringbone	staggered	3.8 %	5310
Serpentine	folded	3.2 %	4870

The herringbone design produced the most complicated field and the largest reconstruction error. Its staggered grooves create counter-rotating vortices that particles cross only briefly. Nevertheless, the main recirculation zones were identified correctly in every clip. We conclude that offline reconstruction is sufficient for routine quality control of fabricated chips.